Integrity & Governance

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Collaborative academic research environments face critical challenges in data trustworthiness, ownership accountability, and duplication of research efforts. This research proposes an **Integrity & Governance Module** that leverages a blockchain-inspired append-only ledger to ensure tamper-evident traceability of research activities. Each lifecycle event—from topic submission to milestone validation—is recorded using cryptographic hashes and timestamps, establishing a verifiable lineage of research artifacts. To prevent intellectual redundancy, the system integrates **Sentence-BERT (SBERT)** for semantic embedding and **FAISS** for high-speed vector similarity search, enabling real-time detection of semantically overlapping research topics. The expected outcomes include a transparent contribution mapping framework, a tamper-proof provenance record, and a scalable mechanism for maintaining academic integrity within collaborative research workflows.

Keywords: Data Provenance, Blockchain Governance, Smart Contracts, Semantic Similarity, SBERT, FAISS, Academic Research Integrity.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ABAC	Attribute-Based Access Control
AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
DAG	Directed Acyclic Graph
DLT	Distributed Ledger Technology
EVM	Ethereum Virtual Machine
FAISS	Facebook AI Similarity Search
IPFS	InterPlanetary File System
KG	Knowledge Graph
NLP	Natural Language Processing
PoA	Proof of Authority
PROV	W3C Provenance Model
SBERT	Sentence-BERT
SOTA	State-of-the-Art
W3C	World Wide Web Consortium

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Appendix	Title
Appendix A	Stakeholder Survey Instrument
Appendix B	Detailed System Architecture and Data Flow Diagrams
Appendix C	Preliminary UI Mockups

1. INTRODUCTION

1.1 Background and Context

Collaborative academic research environments are increasingly data-intensive, yet existing systems struggle to maintain trust, transparency, and accountability. Centralized research management platforms often suffer from single points of failure, fragmented databases, and inconsistent record-keeping, which can compromise the integrity of a project's history. This problem affects multiple stakeholders: **students** may face disputes over intellectual contributions, **supervisors** encounter difficulties in verifying approvals, and **institutions** risk reputational damage due to research misconduct or duplication.

Globally, research duplication and plagiarism remain persistent issues. Studies indicate that a significant proportion **of academic proposals contain overlapping or semantically similar content** with previously submitted work, often undetected by traditional keyword-based tools. Locally, universities struggle to track research milestones efficiently, leading to delayed project approvals and missed contributions.

Existing solutions, such as Overleaf for collaborative writing, Git for version control, and conventional plagiarism detectors, provide partial solutions but fail to deliver **tamper-evident provenance, integrated governance, and semantic novelty detection**. The proposed Integrity & Governance Module addresses these gaps by combining:

- A **blockchain-inspired append-only ledger** for tamper-evident tracking of research events.
- **Semantic similarity detection** using Sentence-BERT (SBERT) and FAISS to prevent idea duplication.
- Transparent **contribution mapping and approval governance**, enabling accountability across the research lifecycle.

This framework also aligns with **UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure)** by promoting secure and innovative research infrastructures, and fits within the **Research Cluster on Academic Governance and Digital Innovation**.

1.2 Literature Review

Recent studies indicate significant advances in blockchain, provenance tracking, and AI-driven semantic analysis:

1. **Blockchain-based provenance** ensures the integrity and lineage of digital artifacts **[1], [4]**. W3C PROV standards categorize artifacts as Entities, Activities, and Agents, enabling detailed tracking of research history. DAG-oriented ledgers allow retrospective verification of every modification **[3]**.

2. **Smart contract-based governance** facilitates automated enforcement of policies for approvals and milestone validations **[11]**. Research demonstrates that Proof-of-Authority (PoA) models and integrity verification frameworks are particularly effective in private academic environments due to lower latency and higher throughput **[10]**.

3. **Semantic similarity detection** with SBERT dramatically improves the detection of semantically overlapping topics, achieving speedups of over 900× compared to traditional BERT models while maintaining high accuracy **[5]**.

4. **AI-based novelty analysis** and hybrid models (SBERT+Transformers, NBSVM) capture long-term semantic dependencies, distinguishing original contributions from paraphrased or derivative content **[6], [7]**.

5. **Knowledge Graphs for academic data** organize heterogeneous research inputs and ensure only high-quality triples contribute to the canonical research record. Blockchain prevents tampering or submission of incorrect knowledge **[9]**.

- Despite these advances, current systems fail to integrate blockchain-based provenance, semantic analysis, and governance into a **single**, **auditable framework [2], [12]**. Commercial solutions and research prototypes address only parts of the workflow: either focusing on version control, plagiarism detection, or milestone approvals.

1.3 Research Gap

The critical gaps in current academic research systems include:

1. **Lack of unified provenance tracking**

- *Current systems:* Most blockchain platforms focus on supply chains, medical records, or generic file tracking.
- *Gap:* They do not model the complete academic research lifecycle, including topic submissions, proposal milestones, supervisor approvals, and revisions.
- *Impact:* Institutions cannot verify who performed what action or when, creating risks of duplication and disputes.
- *Our solution:* The proposed component explicitly logs all research events in a tamper-evident ledger, enabling traceable governance and accountability.

2. **Insufficient accountability and contributor mapping**

- *Current systems:* Some track user IDs for edits, but ignore role evolution (e.g., student → co-supervisor → committee validator).
- *Gap:* There is no structured mapping between user actions and roles over time.
- *Impact:* Responsibility is unclear, making verification and dispute resolution difficult.
- *Our solution:* The system maps contributions and roles over time, ensuring verifiable accountability for every action.

3. **Inadequate tampering detection and verification**

- *Current systems:* Immutability is often assumed, but few systems simulate tampering or validate the integrity of stored histories.
- *Gap:* Without explicit verification, trust in the records is theoretical rather than proven.
- *Impact:* Undetected modifications can undermine academic integrity and erode trust.
- *Our solution:* The component provides tamper-evident proof, demonstrating that all modifications leave verifiable traces.

4. **Limited semantic novelty detection**

- *Current systems:* Research platforms rely on keyword or metadata matching, missing semantically similar topics.
- *Gap:* Redundant research may go unnoticed, causing wasted effort.
- *Impact:* Students and researchers may unknowingly duplicate work, affecting innovation and publication quality.
- *Our solution:* Integration of a semantic similarity engine detects overlapping topics in real-time, preventing redundant research.

5. **Fragmented integration of governance, AI, and blockchain**

- *Current systems:* Governance, AI-driven analysis, and blockchain are often handled in separate tools.
- *Gap:* No unified framework combines provenance tracking, anomaly detection, and workflow governance.
- *Impact:* Institutions must manage multiple systems, increasing complexity and reducing reliability.
- *Our solution:* The proposed architecture unifies these aspects, offering a single platform for transparent, accountable, and intelligent research workflow management.

Research Question:
" How can blockchain-based provenance tracking and smart contract governance improve integrity, transparency, and accountability in academic research workflows?"

2. OBJECTIVE

2.1 Main Objective

Develop a **blockchain-inspired Integrity & Governance Module** that ensures **tamper-evident tracking, transparent accountability, and semantic novelty detection** for collaborative academic research workflows, measurable by the successful implementation of a prototype system that records research lifecycle events and identifies overlapping topics with >90% accuracy.

2.2 Specific Objectives

- **Design a hash-linked, append-only ledger** to record research lifecycle events, including topic submissions, revisions, and milestone approvals.
- **Implement user contribution mapping**, linking each research action to a verified identity to ensure accountability and traceability.
- **Integrate semantic similarity detection** using Sentence-BERT (SBERT) embeddings and FAISS indexing to identify potentially overlapping or duplicate research topics.
- **Develop a DAG-based provenance visualization**, allowing stakeholders to view the complete historical lineage of research artifacts.
- **Validate system performance**, measuring latency, accuracy of similarity detection, and tamper-evidence of the ledger.
- **Evaluate governance effectiveness**, ensuring smart contract-based approval workflows operate securely and correctly in a multi-user academic environment.

3. METHODOLOGY

3.1 Research Approach

This study adopts a **Design Science Research (DSR)** approach, aimed at **designing, developing, and evaluating** the **Integrity & Governance module** for collaborative academic research systems. DSR is appropriate as it allows iterative design of a practical artifact that ensures **tamper-proof tracking, accountability, and semantic uniqueness** in research workflows.

The problem addressed is the **lack of systems capable of guaranteeing both the integrity of research records and semantic uniqueness** of academic submissions. Existing repositories either provide **basic storage and access control** or rely on traditional keyword-based duplication checks, failing to ensure a **fully auditable and tamper-evident provenance chain**.

To tackle this, the proposed system focuses on the **Integrity & Governance module**, which serves as the **core ledger-based component**, while **semantic analysis** acts as a pre-commit and post-commit support process to prevent redundant submissions and verify novelty.

The DSR process for this study follows five stages:

1. **Problem Identification**: Current systems cannot guarantee tamper-proof, accountable research provenance while ensuring intellectual uniqueness.

2. **Objective Definition**: Design an integrity and provenance framework that records all research lifecycle events immutably and checks semantic uniqueness before storing events.

3. **Artifact Design and Development**: Develop the Integrity & Governance module with pre-commit semantic validation using SBERT and FAISS, DAG-based visualization, and smart contract-driven governance.

4. **Demonstration**: Apply the system to simulated and real academic submissions within a university research environment.

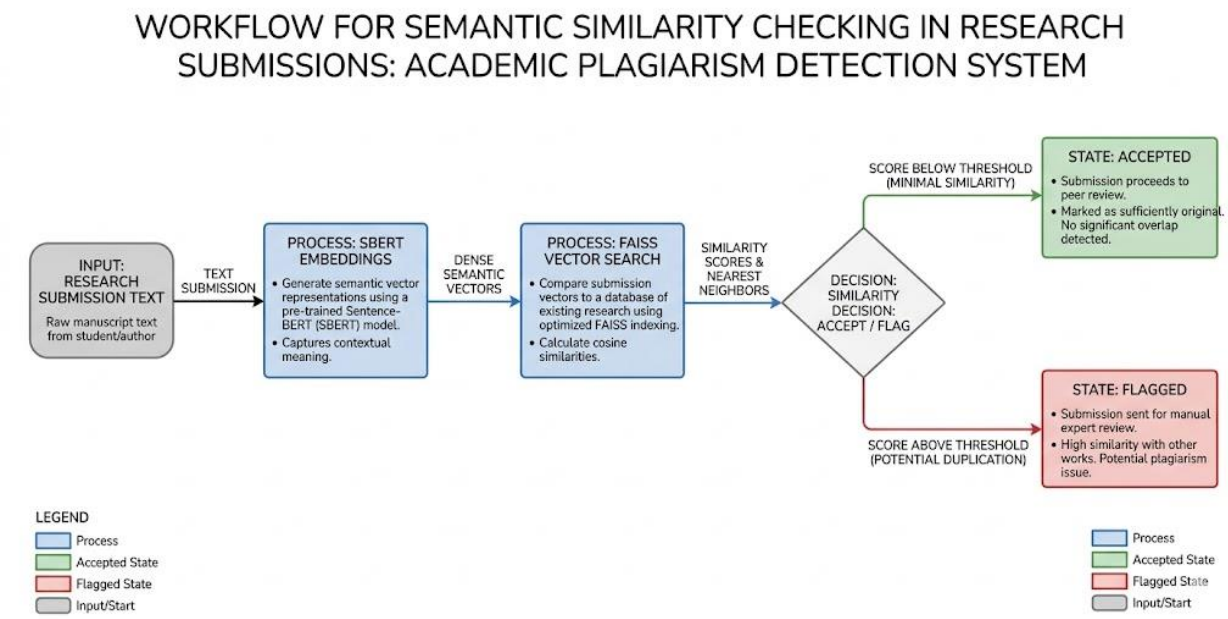
5. **Evaluation**: Validate the module using quantitative metrics (integrity, similarity detection, throughput) and qualitative expert assessments.

3.2 Technical Methods and Algorithms

The **Integrity & Governance module** follows a structured multi-stage workflow:

Stage 1 — Semantic Similarity Detection

- Capture research lifecycle events, including topic submissions, revisions, and milestone approvals.
- Before committing an event to the ledger, perform **semantic uniqueness analysis**:
  - Transform the research title, abstract, and key metadata into **SBERT embeddings**.
  - Query previously stored embeddings using **FAISS** for fast similarity retrieval.
    Reject or flag events that are semantically redundant, preventing duplication of ideas.



Stage 2 — Ledger Event Recording

- Store events in a **hash-linked append-only ledger** with fields: user_id, event_type, previous_block_hash, and timestamp.
- Ensure **tamper-evident provenance**: any modification breaks the chain, allowing instant detection.
- Semantic checks can also be optionally re-run post-commit for auditing purposes.

Stage 3 — Provenance DAG Construction

- Construct a **Directed Acyclic Graph (DAG)** to visualize research lineage, connecting all events, contributors, and their actions.
- Supports **transparent auditing** and traceability of research contributions.

Stage 4 — Governance Enforcement via Smart Contracts

- Implement **Solidity smart contracts** to enforce authorization rules: only designated supervisors or co-supervisors can approve milestones.
- Prevents procrastinating or malicious auditors and ensures **policy compliance** across multi-stakeholder workflows.

Stage 5 — Optional Semantic Audit

- Optionally perform semantic similarity checks on archived submissions using SBERT + FAISS to identify late-stage redundancy or conflicts.
- Enables **continuous quality control** of the academic repository.

Stage 6 — Evaluation and Metrics

- **Quantitative**: ledger integrity verification, pre-commit semantic detection accuracy, throughput and latency under varied transaction loads.
- **Qualitative**: expert review of DAG visualization, governance enforcement, and contribution clarity.

3.3 Tools and Platforms with Justification

- **SBERT**: For semantic embedding of research events; ensures detection of paraphrased or semantically similar submissions.
- **FAISS**: High-speed vector search for real-time semantic uniqueness checks.
- **Solidity / Smart Contracts**: Automates governance, enforcing rules and approvals.
- **DAG Libraries** (e.g., NetworkX or D3.js): Visualize provenance lineage in an interpretable format.
- **IPFS (optional)**: For off-chain storage of research artifacts, keeping the blockchain lightweight while preserving integrity.

3.4 Data Collection and Ethical Considerations

Data Collection

- Open-access academic repositories: arXiv, Semantic Scholar Open Research Corpus, CrossRef metadata.
- Simulated student submissions to represent real academic workflows.

Ethical Considerations

- User submissions are anonymized to protect intellectual property.
- Pre-commit semantic checks do not permanently store sensitive data.
- External services and APIs follow strict zero-data-retention policies.
- All activities comply with institutional ethical guidelines.

3.5 Validation Strategy and Performance Metrics

Quantitative Evaluation

- **Ledger Integrity**: Hash verification ensures tamper-evidence.
- **Semantic Detection**: Precision, Recall, F1-Score, and cosine similarity to measure correctness of pre-commit uniqueness checks.
- **System Performance**: Throughput and latency measurements for real-time event processing.

Qualitative Evaluation

- Expert panels evaluate DAG visualization, governance compliance, and contribution mapping.
- Experts assess clarity, interpretability, and practical usefulness of audit outputs.
- Panel review identifies potential system gaps or misclassifications in semantic uniqueness assessment.

HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution

The proposed system is a **Blockchain-Inspired Integrity & Governance Framework** for collaborative academic research. Its primary objective is to ensure:

- **Tamper-proof provenance** of research lifecycle events
- **Semantic uniqueness detection** to prevent redundant submissions
- **Transparent contribution mapping** for accountability
- **Governance enforcement** using smart contracts

The system revolves around the **Integrity & Governance module**, which acts as the central ledger, with semantic validation as a supporting submodule for both **pre-commit** (before storing an event) and optionally **post-commit** (for audits).

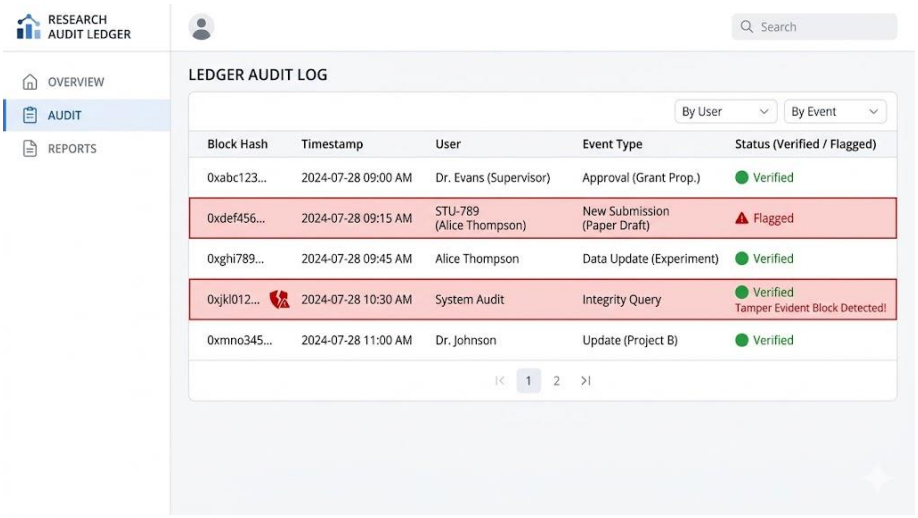
4.2 System Components

1. Integrity & Governance Module (Core)

- **Ledger:** Hash-linked append-only storage of all research events
- **Provenance DAG:** Visual lineage representation of research artifacts and contributors
- **Semantic Validation Submodule:** Uses SBERT + FAISS to detect redundancy before committing an event
- **Smart Contract Engine:** Enforces governance rules and validates supervisor/co-supervisor approvals

2. User Interface / Client Layer

- Submission portal for students
- Dashboard for supervisors to approve milestones
  Visualizations of research provenance and contribution mapping



3. Data Storage Layer

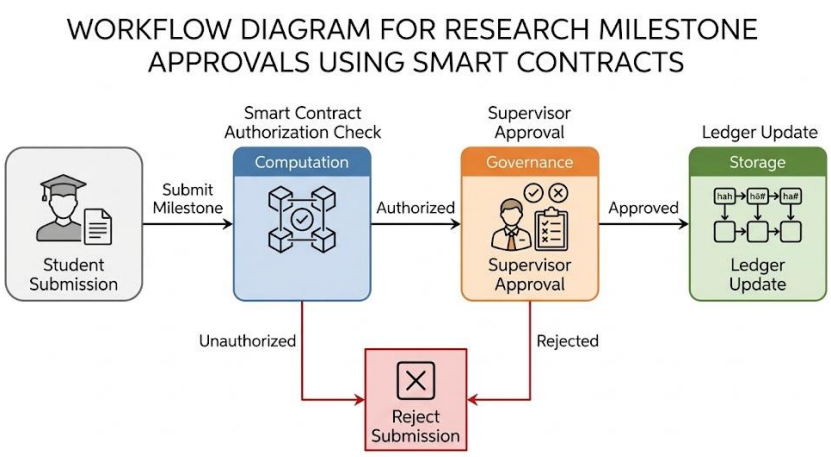
- On-chain: Ledger hashes and metadata
- Optional off-chain storage: Full documents/artifacts (e.g., IPFS)

4. External Services / APIs

- Semantic similarity API (SBERT + FAISS)
- Optional LLM for advanced audits or semantic reporting

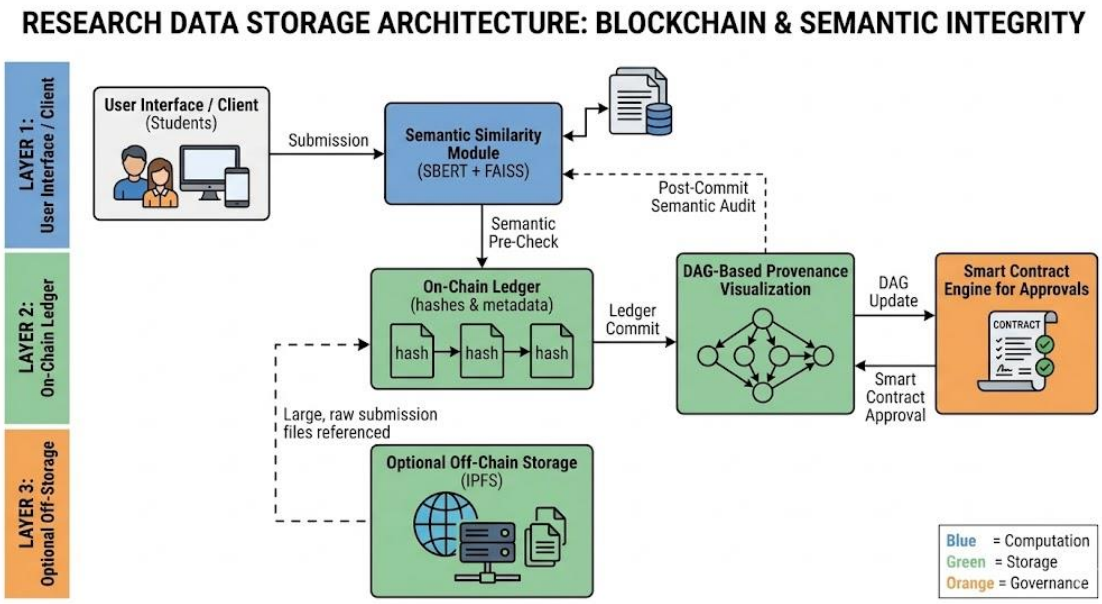
4.3 Component Interactions

1. **Submission Flow:**
    a. A user submits a research event (topic submission, milestone update).
    b. Event is pre-checked by **semantic validation** for duplication.
    c. If unique, event is hashed and committed to the **ledger**.
    d. DAG is updated to reflect new event lineage.
2. **Approval Flow:**
    a. Supervisors/co-supervisors validate milestones via **smart contracts**. Smart contracts enforce authorization and record approvals in the ledger.



3. **Audit / Post-Commit Flow:**
    a. Optional semantic check reruns to detect late-stage redundancy.
    b. DAG visualizations and provenance reports are generated for audits.

4.4 Data Flow Explanation



- Each node in the DAG represents an **event**, linked via **hashes** for immutability.
- Metadata (user ID, timestamp, event type) ensures **traceability**.
- Semantic vectors are stored temporarily for **real-time similarity queries**.

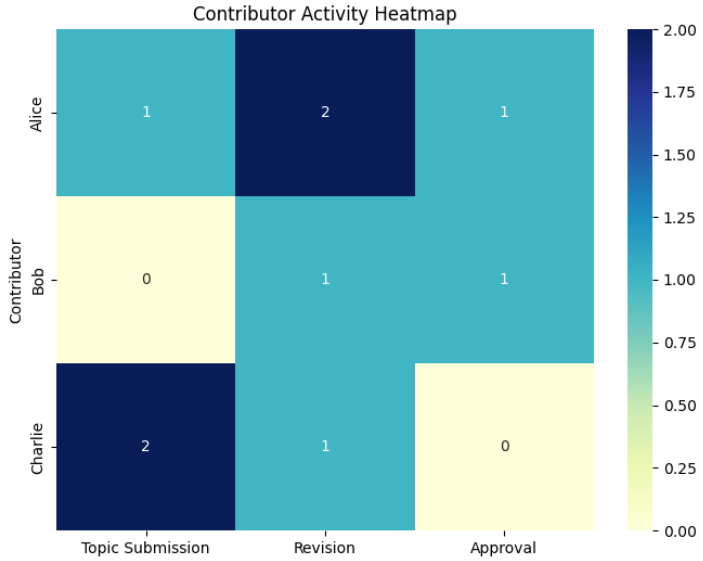
4.5 Scalability and Security Considerations

- **Scalability:**

    - FAISS indexing ensures sub-second semantic retrieval even for **large archives of submissions**.
    - DAG and hash-linked ledger allow horizontal scaling across multiple servers or clusters.
- **Security:**
    - Cryptographic hashes ensure **tamper-evidence** for all events.
    - Smart contracts prevent **unauthorized approvals** or malicious updates.
    - Access control and anonymization maintain **privacy of submissions** while ensuring accountability.
    - Off-chain storage is optional, reducing blockchain bloat without sacrificing integrity

USER REQUIREMENTS

- Stakeholder identification
- Functional requirements (numbered)
- Non-functional requirements (security, performance, usability, etc.)



5.1 Stakeholder Identification

- **Students / Research Contributors**
    a. Submit research topics, updates, and milestones
    b. Access provenance reports and feedback on submission uniqueness
- **Supervisors / Co-Supervisors**
    a. Validate submissions and milestones
    b. Monitor research lineage and contributor activity
- **University / Research Institution Administrators**
    a. Enforce governance policies via smart contracts
    b. Audit provenance records and ensure compliance
- **System Administrators**
    a. Maintain system operations, storage, and security
    b. Manage user accounts and access control

5.2 Functional Requirements

1. The system shall record every research lifecycle event (submission, approval, revision) in a tamper-evident ledger.

2. The system shall enforce supervisor/co-supervisor approvals using smart contracts.

3. The system shall provide real-time semantic similarity analysis to detect redundant research topics before committing an event.

4. The system shall generate a visual DAG representing the full provenance and contribution lineage of research artifacts.

5. The system shall allow querying historical records by user, topic, timestamp, or semantic similarity.

6. The system shall produce audit reports summarizing provenance, semantic uniqueness, and contribution mapping.

5.3 Non-Functional Requirements

- **Security:**
    a. All events must be cryptographically hashed before storage
    b. Smart contracts enforce role-based access and prevent unauthorized modifications
- **Performance:**
    a. Semantic similarity checks shall return results in <2 seconds for large research archives
    b. Ledger commits and DAG updates shall handle concurrent submissions efficiently
- **Usability:**
    a. User-friendly dashboards for students, supervisors, and administrators
    b. Clear provenance visualizations and alerts for redundant or flagged submissions
- **Scalability:**
    a. The system shall handle increasing numbers of submissions and users without degradation in response time
- **Reliability & Integrity:**
    a. Tamper-evident ledger ensures all events are immutable
    b. Continuous verification mechanisms detect broken or modified records

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

- **Target Market:** Higher education institutions, research universities, and corporate R&D departments.
- **Customer Persona:**
  .1. **University IT / Research Office Managers:** Seek systems that ensure research integrity, prevent redundant work, and provide transparent provenance tracking.
  .2. **Supervisors / Professors:** Require tools to monitor student contributions and validate milestones efficiently.
  .3. **Students / Researchers:** Need assurance that their contributions are protected and properly credited.

6.2 Value Proposition

- Provides **tamper-evident provenance tracking** for research activities.
- Detects **redundant or overlapping research topics** in real-time using semantic analysis.
- Automates enforcement of governance policies with **smart contracts**, reducing administrative workload.
- Enhances **academic credibility** and reduces intellectual property disputes.

6.3 Revenue Model

- **Software-as-a-Service (SaaS):** Subscription-based access for institutions (monthly or annual).
- **License Fee:** One-time installation fee for private on-premise deployment.
- **Add-on Modules:** Optional advanced analytics, semantic plagiarism reports, or custom dashboards.

6.4 Cost Estimation

- **Development Costs:** Implementation of ledger, semantic modules, dashboards (~60,000 LKR).
- **Infrastructure Costs:** Cloud hosting for prototype deployment, storage for test datasets, and API hosting. (~20,000 LKR).
- **Maintenance & Testing:** System testing, dataset preparation, and evaluation environment. (~10,000 LKR)
- **Miscellaneous Costs:** Document Preparation, backup storage, and experimental resources. (~10,000 LKR).

6.5 Pricing Strategy

- **Tiered Pricing:**
  .1. Small institutions: 50,000 LKR per month
  .2. Medium institutions: 120,000 LKR per month
  .3. Large universities: 250,000 LKR per month

- **Add-ons:** Premium semantic analytics or provenance visualization charged separately.
- Discounted rates for long-term contracts or academic consortia.

6.6 Competitive Advantage

- **Blockchain-inspired integrity:** Unlike traditional academic management systems, all records are immutable.
- **Semantic novelty detection:** Identifies overlapping topics even with paraphrased submissions.
- **Integrated governance:** Combines provenance, accountability, and semantic analysis in one unified system.
- **Scalable & privacy-conscious:** Designed for increasing volumes of research while protecting contributor IP.

6.7 Intellectual Property Considerations

- Core algorithms (ledger structure, semantic scoring, DAG visualization) will be **patentable software innovations**.
- Source code protected under **copyright law**; licensing ensures institutions cannot redistribute the software.
- Semantic embeddings and models leverage **open-source AI frameworks**, while custom modules maintain IP ownership.

BUDGET AND JUSTIFICATION

7.1 Hardware Costs

- **Developer Laptops / Workstations:** Already owned by the team.
  Justification: Development, semantic model training, and testing will be done on existing hardware. No new purchase is required.

7.2 Software Licenses

Software Tool	Purpose	License Cost
Python	Programming language	Free
PyTorch	Deep learning framework	Free
HuggingFace Transformers	NLP model implementation	Free
FAISS	Vector similarity search	Free
FastAPI	Backend API development	Free
Solidity & Web3.py	Smart contract development	Free
Draw.io / Diagrams.net	System architecture diagrams	Free

7.3 Cloud Services

Cloud Service	Purpose	Estimated Cost (LKR)
Google Colab / Kaggle	Model training and experimentation	Free
GitHub Actions	CI/CD automation	Free
Render / Railway Free Tier	Backend deployment for prototype	Free tier
Cloud Storage (optional testing)	Temporary dataset storage	15,000
Backup Storage	Experimental data backup	5,000
Total		20,000

7.4 Human Effort Cost

Role	Task	Estimated Cost (LKR)
Student Researcher / Developer	System design, development, semantic model integration	60,000
Testing & Validation Support	System testing and evaluation	10,000
Documentation & Experiments	Dataset preparation and reports	10,000
Total		80,000

Justification: The main investment is team effort. Using free tools reduces financial costs to near zero.

Grand Total Estimated Budget: 100,000 LKR

Human Effort Justification

Although the project is conducted as an academic research study, a small estimated value is assigned to represent the development and testing effort invested during the research process.

7.5 Overall Budget Summary

Budget Category	Estimated Cost (LKR)
Hardware Costs	0
Software Licenses	0
Cloud Services	20,000
Human Effort Cost	80,000
Total Estimated Budget	100,000

WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown Structure

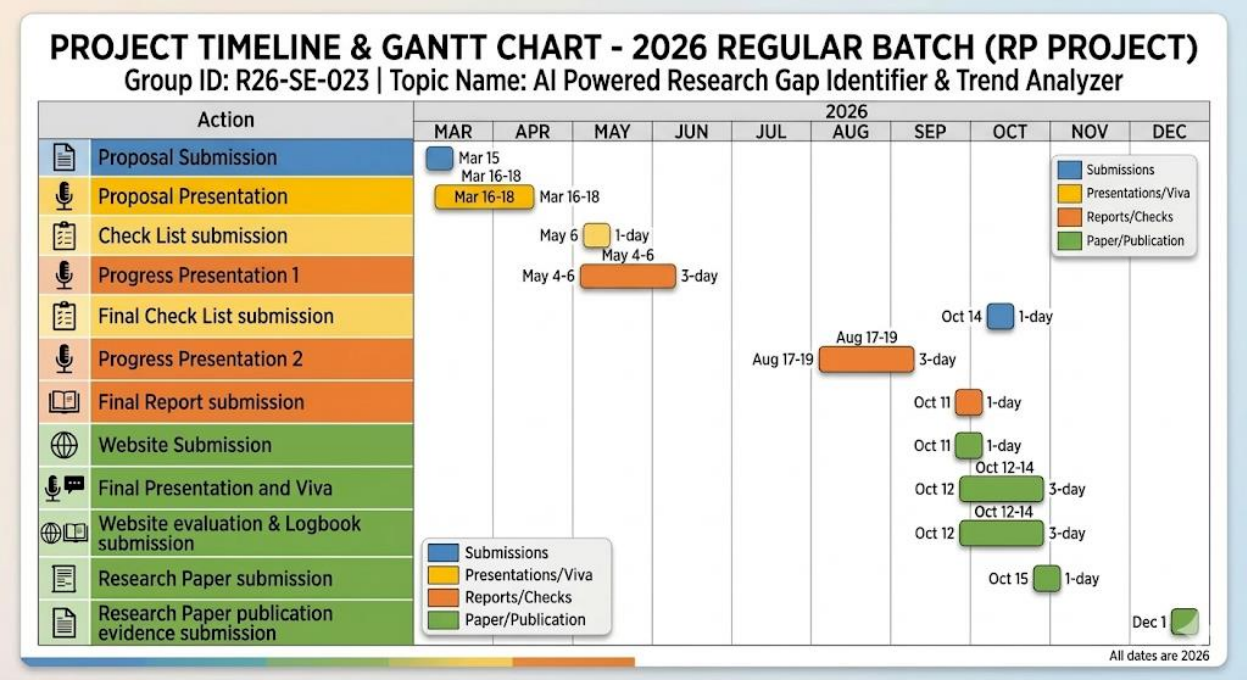
Task ID	Task Name	Description	Timeline
T1	Literature Review	Review research papers on blockchain-based data provenance, smart contract security, and semantic similarity techniques used in research integrity systems.	March 2026
T2	Requirements Analysis	Identify functional and non-functional requirements for the research integrity and provenance module including topic submission, approval tracking, and revision logging.	March 2026
T3	System Architecture Design	Design the high-level architecture including the integrity ledger, semantic similarity module, and integration with the collaborative research workflow system.	March – April 2026
T4	Ledger Data Structure Design	Design the blockchain-inspired append-only ledger structure including block format, hash linking, timestamping, and contributor metadata.	April 2026
T5	Provenance Recording Module	Implement the provenance recording mechanism to capture research events such as topic submission, revision, and supervisor approvals.	April – May 2026
T6	Semantic Similarity Module	Implement SBERT-based semantic embedding generation to detect similarity between proposed research topics and existing projects.	May 2026
T7	Vector Index Integration	Integrate FAISS vector search to enable efficient semantic similarity retrieval from the research topic database.	May – June 2026
T8	Integrity Verification Engine	Develop verification algorithms that validate ledger entries using hash comparisons to detect tampering or modification of research records.	June 2026
T9	Workflow Integration	Integrate the integrity and provenance module with the collaborative research management system including submission, approval, and revision processes.	June – July 2026
T10	Visualization Dashboard	Develop a dashboard to display research lineage, provenance graphs, and similarity scores for supervisors and administrators.	July 2026

T11	System Testing and Validation	Evaluate system performance including semantic similarity accuracy, ledger integrity verification, and workflow traceability.	August 2026
T12	Documentation and Final Report	Prepare system documentation, diagrams, experimental results, and final research report for submission and presentation.	September 2026

8.2 Timeline Alignment

Phase	Period	Activities
Phase 1	March 2026	Literature review and requirements analysis
Phase 2	March – April 2026	System architecture design and ledger structure design
Phase 3	April – May 2026	Provenance recording module development
Phase 4	May – June 2026	Semantic similarity module and FAISS vector search integration
Phase 5	June – July 2026	Integrity verification engine and workflow integration
Phase 6	July 2026	Visualization dashboard development
Phase 7	August 2026	System testing and performance evaluation
Phase 8	September 2026	Documentation, final validation, and proposal submission

GANTT CHART


PROJECT TIMELINE & GANTT CHART - 2026 REGULAR BATCH (RP PROJECT)
Group ID: R26-SE-023 | Topic Name: AI Powered Research Gap Identifier & Trend Analyzer
Action
2026
MAR
APR
MAY
JUN
JUL
AUG
SEP
OCT
NOV
DEC
Proposal Submission
Proposal Presentation
Check List submission
Progress Presentation 1
Final Check List submission
Progress Presentation 2
Final Report submission
Website Submission
Final Presentation and Viva
Website evaluation & Logbook submission
Research Paper submission
Research Paper publication evidence submission
Mar 15
Mar 16-18
Mar 16-18
Mar 16-18
May 6
1-day
May 4-6
May 4-6
3-day
Oct 14
1-day
Aug 17-19
Aug 17-19
3-day
Oct 11
1-day
Oct 11
1-day
Oct 12-14
Oct 12
3-day
Oct 12-14
Oct 12
3-day
Oct 15
1-day
Dec 1
Submissions
Presentations/Viva
Reports/Checks
Paper/Publication
Submissions
Presentations/Viva
Reports/Checks
Paper/Publication
All dates are 2026

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APPENDICES

Appendix A – Stakeholder Survey Instrument

A preliminary survey was conducted among students and academic staff to understand challenges in research topic management and revision tracking.

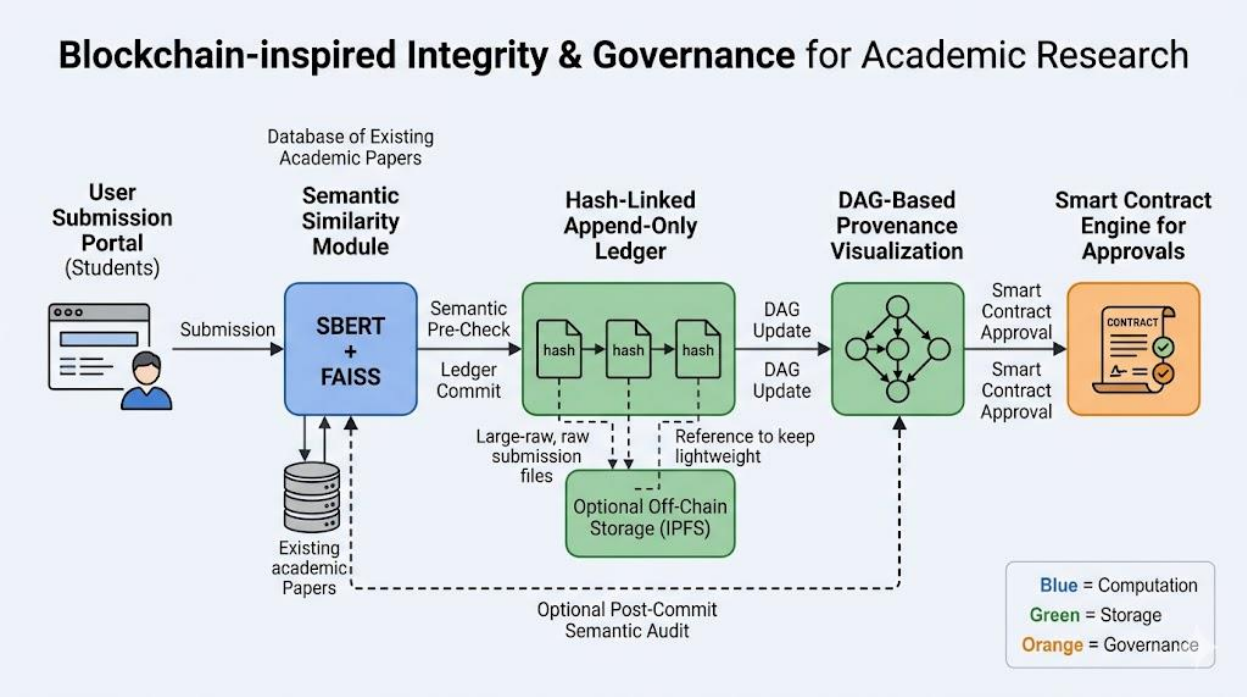
Sample questions:

1. How does your institution currently verify whether a research topic already exists or is too similar to previous work?
2. How confident are you that current research project management systems maintain a **tamper-proof revision history**?
   (1 = Not confident, 10 = Very confident)
3. What challenges do supervisors face when tracking research topic revisions and approvals?

Appendix B – Detailed System Diagrams

This appendix contains the following diagrams:

- High-Level System Architecture Diagram
- Data Flow Diagram (Research submission → Semantic similarity check → Ledger recording)
- Integrity Ledger Structure showing **hash-linked research events**



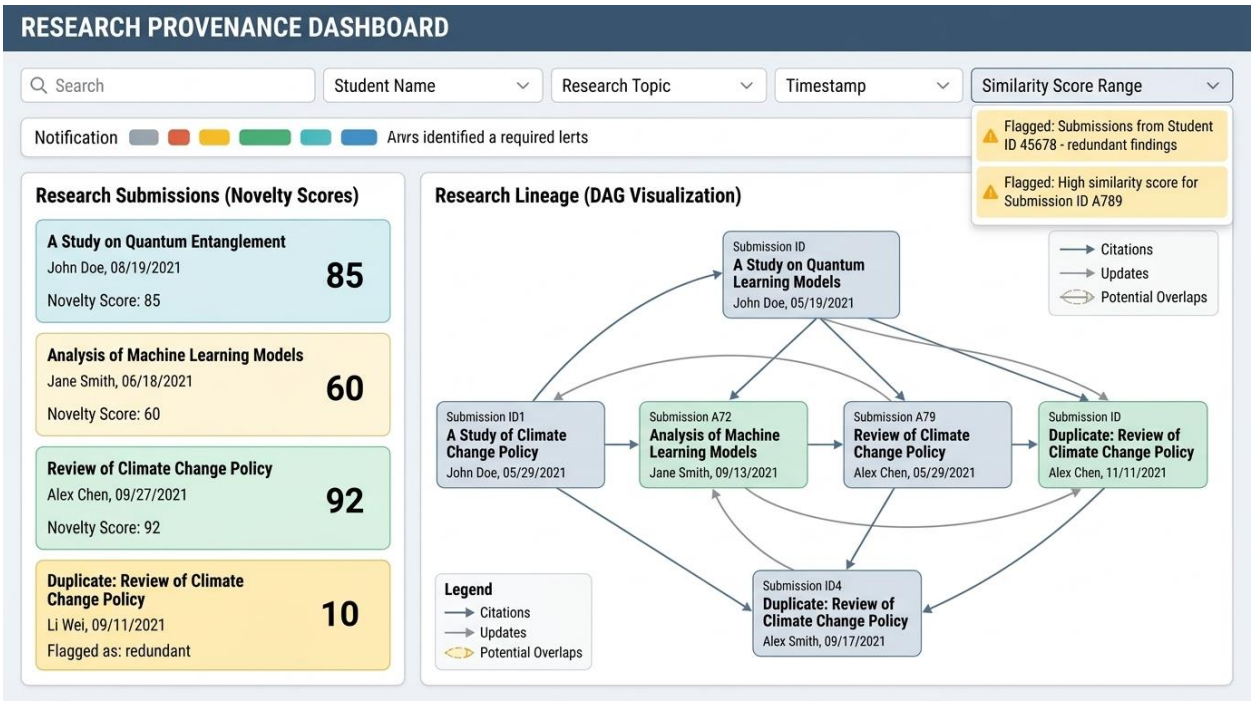
Appendix C – Preliminary UI Mockups

Early prototype interfaces include:

• **Research Topic Submission Interface** displaying novelty score generated by semantic similarity analysis.

• **Research Provenance Dashboard** showing approval workflow and revision history.

• **Integrity Audit View** allowing administrators to verify the blockchain-style ledger entries.


RESEARCH PROVENANCE DASHBOARD
Search
Student Name
Research Topic
Timestamp
Similarity Score Range
Notification
Arvrs identified a required lerts
Flagged: Submissions from Student ID 45678 - redundant findings
Flagged: High similarity score for Submission ID A789
Research Submissions (Novelty Scores)
A Study on Quantum Entanglement
John Doe, 08/19/2021
Novelty Score: 85
85
Analysis of Machine Learning Models
Jane Smith, 06/18/2021
Novelty Score: 60
60
Review of Climate Change Policy
Alex Chen, 09/27/2021
Novelty Score: 92
92
Duplicate: Review of Climate Change Policy
Li Wei, 09/11/2021
Flagged as: redundant
10
Research Lineage (DAG Visualization)
Citations
Updates
Potential Overlaps
Submission ID
A Study on Quantum Learning Models
John Doe, 05/19/2021
Submission ID1
A Study of Climate Change Policy
John Doe, 05/29/2021
Submission A72
Analysis of Machine Learning Models
Jane Smith, 09/13/2021
Submission A79
Review of Climate Change Policy
Alex Chen, 05/29/2021
Submission ID
Duplicate: Review of Climate Change Policy
Alex Chen, 11/11/2021
Submission ID4
Duplicate: Review of Climate Change Policy
Alex Smith, 09/17/2021
Legend
Citations
Updates
Potential Overlaps